

CPS 150 Review Session #2

Java Variables and Strings

Review Activity

What is the output of this program?

```
public class Review {  
    public static void main(String[] args) {  
        int var1 = 9;  
        int var2 = 2;  
        int var3 = var1 / var2;  
        int var4 = var3 % 3;  
  
        System.out.println("The result is");  
        System.out.print(var3);  
        System.out.print(" --- ");  
        System.out.println(var4);  
    }  
}
```

Type Casting

Allows us to convert from one type to another

```
int num1 = 5;  
int num2 = 2;  
double div1 = num1 / num2; // 2.0  
double div2 = (double) num1 / (double) num2; // 2.5
```

Important Use of Type Casting

Random numbers with `Math.random()`

```
int randNum1 = (int) (Math.random() * 100) + 1; // rand num 1-100
```

Alternative Using Random Class

```
Random rand = new Random();  
int randNum2 = rand.nextInt(100) + 1;
```

Characters

`char` is used to store an individual character/symbol

Escape Sequences

Special characters that allow for specific formatting

Sequence	Use
<code>\n</code>	New line
<code>\t</code>	Tab
<code>\"</code>	Quotation mark

Strings

Strings are an object in Java that store a sequence of characters

```
String myString = "Hello World";
```

Objects vs Primitives

Strings are different than primitive data types (`int`, `double`, etc.) because they have special properties (methods) built in with them, and they store data uniquely

String Methods

Method	Result
<code>myString.toUpperCase()</code>	<code>HELLO WORLD</code>
<code>myString.toLowerCase()</code>	<code>hello world</code>
<code>myString.length()</code>	<code>11</code>
<code>myString.substring(3, 8)</code>	<code>lo Wo</code>
<code>.charAt(8)</code>	<code>r</code>
<code>.replace('l', 'p')</code>	<code>Heppo Worpd</code>

Strings with Scanner

Two ways to read Strings with a Scanner

```
Scanner scan = new Scanner(System.in);  
String input1 = scan.next(); // will read in next word up to a space  
String input2 = scan.nextLine(); // will read in all words up to new line
```

Clearing input

Use a `.nextLine()` that does not save the data to clear out your Scanner when getting multiple types of input

Practice Problem

Write a program to print a formatted movie ticket receipt at a theater where adult tickets are \$10.99 and kids tickets are \$6.99

Example Run

```
Enter your name: Bob
Enter your movie name: Spider-Man
How many adult tickets: 2
How many kids tickets: 2
Your receipt:
Spider-Man --- Bob
Adult Tickets x2 - $21.98
Kids Tickets x2 - $13.98
Total - $35.96
```